

Exponential Expressions

Teacher's Edition

Read This First

This lesson rebuilds Math Nation 3.3 (SE pp. 112–117), the book's 'apply more than one law' day. The book's three collaborative pieces — Winston/Yvette, Caroline/Jamir, and the roundtable in teams of three — are here as Notes II, Example 1 and the whiteboard round. Nothing mathematical in the book's lesson pages needed repair.

One repair on the Additional bank: the book's Homework L3 #7 key prints $(\frac{3}{4})^6 = 729/4,096$; the denominator is $4^6 = 4,096$.

Notes I is a seven-row reference table. Under ruling 11 the student's notes are the only durable record of the unit, so this table is copied in full and the teacher's edition says so.

Standard and Targets

MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational-number bases.

Clarification Two or three laws in one expression, still with whole-number exponents and rational bases. Combined-law work with negative exponents is 3.06–3.07.

Boundary The laws never cross addition or subtraction; Notes III says so and the warm-up's order-of-operations item sets it up.

Learning target I can apply more than one law of exponents to evaluate an expression and to write an equivalent one.

Essential question How can the laws of exponents be used to generate equivalent numeric expressions?

Building on Lessons 3.01–3.02 — the seven laws; MA.7.NSO.2.1 — order of operations with rational numbers.

Working toward MA.8.NSO.1.3 — the same combined work with integer exponents (3.06–3.07); MA.8.AR.1.1 — with variable bases (Thread A).

Where this sits in the course Last of the three MA.7.NSO.1.1 lessons; 3.04 introduces negative exponents (MA.8.NSO.1.3).

Lesson at a Glance

Windy Hill periods run **53 minutes**. The whiteboard round takes the remainder after the taught blocks — **20 minutes** in this lesson — and IXL takes the last 5. Slide numbers below are read from the deck when this edition is built.

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2-3	5 min
Notes	4-6	10 min
Example 1	7-8	3 min
Your Turn	9-10	3 min
Example 2	11-12	3 min
Your Turn	13-14	3 min
Whiteboards	15-32	20 min
IXL	33	5 min
Total		53 min

Materials Whiteboards and markers. Calculators are allowed on everything; the decimal values in the bank (5.0625, 2,293.45007, 9,605.9601) are calculator work once the single power is found.

Where the lesson lives Notes II — two plans, one value. If students leave with 'the order is mine to choose and the value does not change', the lesson worked. Whiteboard 5 packs four named errors into one expression and is the round's best diagnostic.

No partner or group work anywhere in this lesson. Every Math Nation exploration this lesson draws on was written for pairs; here the same tables are worked from the front and on individual boards.

Vocabulary

equivalent expressions — two expressions with the same value for every choice of numbers; the laws of exponents turn one into the other

evaluate — find the single value of an expression

plan — the order in which you will apply the laws; any correct order gives the same value

Warm-Up — spaced retrieval

Four questions, two minutes silent, then the reveal. No reteach longer than one sentence per question.

1. Find the value: $\frac{10^5}{10^3}$ **100**
retrieves: yesterday — 3.02 — quotient of powers

2. Is $\sqrt{50}$ closer to 7 or to 8? **7**
retrieves: last week — Unit 2 — approximating roots (2.04): 50 is 1 above 49 and 14 below 64

3. Solve $4(x-2)=12$ **$x=5$**
retrieves: last unit — Unit 1 — equations with parentheses (1.05)

4. Evaluate $2 + 3 \cdot 4^2$ **50**

*retrieves: prior grade — the prerequisite this lesson needs — MA.6.NSO / 7.NSO.2.1 —
order of operations: the exponent first, then multiply, then add*

Slide-by-Slide Teaching Notes

Slide 1. Exponential Expressions 1 min

- Post the learning target. Say the 'today' line and nothing else yet.

Slide 2. Warm-Up — On your own. Four minutes. No notes. 4 min

- Q4 is the prerequisite: exponents before multiplication before addition. A board with 80 (added first) or 146 (multiplied $3 \cdot 4$ first) is the order-of-operations slip that will show up again inside today's expressions.

Slide 3. Warm-Up — Answers. 1 min

- Reveal. Ask for the band each question came from only if time allows.

Slide 4. Notes — Copy the table. It is your reference for the rest of the unit. 3 min

- This table is the durable record: students have no worksheet and no study guide, so the notes are what they keep (ruling 11). Read down the middle column once, aloud. Then the one sentence that organises the table: product and quotient laws need the SAME base; the power laws move an exponent onto every factor or exponent inside.

Slide 5. Notes — Two students, two orders, one value. 4 min

- Winston: quotient first (inside the parentheses), then power of a power. Yvette: power of a quotient first, then power of a power on top and bottom, then quotient. Both reach $2^6 = 64$. The teaching point is that a plan is required and the order is free — not that one order is right.
- **OFF THE SLIDE, YOURS TO SAY:** 'Pick the order that makes the numbers smallest soonest. Inside the parentheses first is usually that order.'

Slide 6. Notes 3 min

- The laws are shortcuts for multiplying and dividing powers. They never cross an addition or subtraction: $2^3 + 2^2$ is $8 + 4 = 12$, not 2^5 . Parentheses first, then exponents (that is where the laws live), then multiply and divide left to right, then add and subtract.

Slide 7. Example 1 — A zero exponent inside a product inside a quotient. 1 min

- SPOKEN ONLY. Ask for a plan out loud before anything is written: what will you do with 4^0 ? What law joins 4^5 and 4^0 ? What law handles the fraction bar?

Slide 8. Example 1 — Worked — Caroline's plan and Jamir's plan. 2 min

- Caroline adds $5 + 0$ first (product of powers), then subtracts. Jamir replaces 4^0 with 1 first (zero exponent), multiplies, then subtracts. Same $4^3 = 64$. Ask which plan they prefer — the answer does not matter, having one does.

Slide 9. Your Turn — Same steps, your numbers. Boards up when done.

2 min

- Catches a student who reads 5^0 as 0 (the whole product becomes 0) and one who adds $4 + 0 + 2$ (5^6). 25 is right.

Slide 10. Your Turn — Answer. *1 min*

- Reveal; name what a wrong board most likely did (see the note above).

Slide 11. Example 2 — Three laws, a decimal and a negative base. *1 min*

- SPOKEN ONLY. Plan: 11.6^0 is 1. The fraction has two different bases with the same exponent — that is the power of a quotient read backwards, so $(-15 \div 1.5)^4$. Ask what $-15 \div 1.5$ is before anyone raises anything.

Slide 12. Example 2 — Worked. *2 min*

- Zero exponent, then power of a quotient backwards, then one power of -10 . Four negative factors: positive. A student who leaves 11.6 in the answer never applied the zero law; one who gets -10000 miscounted the sign.

Slide 13. Your Turn — Same steps, your numbers. Boards up when done. *2 min*

- Catches a dropped sign (27) and a student who applied the zero law to the wrong power ($7 \cdot$ something). -27 is right.

Slide 14. Your Turn — Answer. *1 min*

- Reveal; name what a wrong board most likely did (see the note above).

Slide 15. Whiteboards · Question 1 of 9 *20 min for the round*

- Example 1's shape with a new base. A board with 0 read 6^0 as 0; a board with 1296 forgot the denominator.

Slide 16. Whiteboards · Question 1 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 17. Whiteboards · Question 2 of 9

- Product then power of a power. A board with 27 used 3^3 (added $0 + 2 + \dots$ then lost the multiply); $3^5 = 243$ added $2 + 3$.

Slide 18. Whiteboards · Question 2 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 19. Whiteboards · Question 3 of 9

- Power of a power then quotient. $0.5^4 = 1/16$. A board with 0.125 stopped at 0.5^3 (divided 6 by 2).

Slide 20. Whiteboards · Question 3 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 21. Whiteboards · Question 4 of 9

- Identity exponent on the outside; quotient inside. $2.5^2 = 6.25$. A board with 2.5 subtracted to 1.

Slide 22. Whiteboards · Question 4 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 23. Whiteboards · Question 5 of 9

- Diagnostic. Four errors live in this one expression; each option is one of them. A board full of C is the zero-exponent misconception again — it should be dying by now.

Slide 24. Whiteboards · Question 5 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 25. Whiteboards · Question 6 of 9

- Diagnostic; it is the book's own ticket-out-the-door item, kept because it is good. The route is power of a quotient backwards: $(0.2 \div 2)^3 = 0.1^3$, then $0.1^3 \cdot 0.1^2 = 0.1^5 = 0.00001$.

Slide 26. Whiteboards · Question 6 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 27. Whiteboards · Question 7 of 9

- Identity, power of a power, product. Four decimal places. A board with 0.001 has 0.1^3 from adding $2 + 1$.

Slide 28. Whiteboards · Question 7 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 29. Whiteboards · Question 8 of 9

- Same base throughout. Three negative factors: negative. A board with 64 dropped the sign; a board reading 'undefined' read $(-4)^0$ as 0 and divided by it — a good error to name aloud.

Slide 30. Whiteboards · Question 8 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 31. Whiteboards · Question 9 of 9

- Written. Full credit needs the two law names (Winston: quotient of powers; Yvette: power of a quotient / power of a power) and the value. A student who says 'no, because the expressions look different' has not yet accepted that equivalent means same value.

Slide 32. Whiteboards · Question 9 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 33. IXL — Last five minutes. 5 min

- IXL: whatever is not finished is tonight's practice; SmartScore 67.

The Whiteboard Round — what each board tells you

Nine questions, one at a time, each with its reveal. Boards down until the cue, all up together, scan the back row first, question the blank boards before the wrong ones. Question 9 is written work. Every wrong option on a multiple-choice question is a named error; the benchmark it comes from is cited.

#	Question	Answer	What a wrong board says
1	$\frac{6^4 \cdot 6^0}{6^2}$	36	0 — zero exponent read as 0 [B1G-M 8.NSO.1.3 misconception]; 1296 — the 6^2 in the denominator was ignored
2	$(3^0 \cdot 3^2)^3$	729	243 — added the exponents $2 + 3$ in the power of a power step [confused with product of powers]; 0 — zero exponent read as 0
3	$\frac{(0.5^3)^2}{(0.5)^2}$	0.0625	0.125 — divided the exponents $6 \div 2$ [7.NSO.1.1 misconception: 'divide' applied to the exponents]; 0.015625 — added $6 + \dots$ or used 0.5^6 and ignored the denominator
4	$\left(\frac{2.5^5}{2.5^3}\right)^1$	6.25	2.5 — the outside 1 was treated as the final exponent [identity law misapplied]; 15.625 — used 2.5^3
5	Which is the value?	A — 4	(B) added the exponents $12 + 10$ instead of subtracting [confused with product of powers]; (C) 3.2^0 read as 0, so the whole numerator is 0 [B1G-M 8.NSO.1.3 zero-exponent misconception]; (D) $(2^2)^6$ written as 2^8 by adding $2 + 6$, then $2^8 \div 2^{10}$ [power of a power confused with product of powers]
6	Which expression is equivalent?	A — 0.1^5	(B) quotient of powers applied to DIFFERENT bases, 0.2 and 2 [7.NSO.1.1: the quotient law needs the same base]; (C) multiplied the exponents $3 \cdot 2$ [confused with power of a power]; (D) the 2^3 in the denominator was ignored [order of operations: the fraction bar is a grouping]
7	$0.1^2 \cdot (0.1^1)^2$	0.0001	0.001 — 0.1^3 from adding $2 + 1$ (power of a power step done as a product); 0.01 — stopped at 0.1^2
8	$\frac{(-4)^2 \cdot (-4)^1}{(-4)^0}$	-64	64 — dropped the sign; 'undefined' or 0 — $(-4)^0$ read as 0 [B1G-M 8.NSO.1.3]

#	Question	Answer	What a wrong board says
9	Winston writes $(2^6/2^4)^3 = (2^2)^3$. Yvette writes $(2^6/2^4)^3 = 2^{18}/2^{12}$. Do they get the same value? Explain.	Yes — both give $2^6 = 64$	'No' — judged equivalence by appearance instead of value [7.NSO.1.1: generate equivalent expressions]

Question Bank — variation and answers

Variation. Question 1 is a read-the-work item (name the laws in two students' plans). Questions 2–5 hold the shape 'two laws, evaluate' and vary the second law: power of a power, then zero exponent, then identity, then a zero exponent in a denominator. 6–7 add a third law. The pattern breaks at 8, a select-all where three of six options are equivalent for three different reasons, and at 11, where the exponent is unknown inside a two-law expression. The Additional sheet mirrors every position with different numbers.

The bank and its Additional sheet are posted, never assigned; they feed the quizzes. Answers below are the same values the keys print — both come from one source.

Question Bank

- (a) zero exponent, then power of a power (b) power of a product (the exponent 4 went to both factors) (c) quotient of powers; the value is 5**
- $(1.5^2)^2 = 1.5^4 = 5.0625$
- $(1 \cdot \frac{4}{5})^2 = \frac{16}{25}$
- $4.7^2 \cdot 4.7^3 = 4.7^5 = 2,293.45007$
- $\frac{9.9}{1} \cdot 9.9^3 = 9.9^4 = 9,605.9601$
- $\frac{2^{12}}{2^{10}} = 2^2 = 4$
- $\frac{(-4)^3}{1} = -64$
- A, C, E A, C and E**
- A A**
- (a) $n = 4$ (b) $n = 5$**
- Yes — power of a quotient, read from right to left; $(-10)^4 = 10,000$.**

Question Bank — Additional

- (a) quotient of powers, then power of a power (b) power of a quotient (with power of a power on top and bottom), then quotient of powers (c) zero exponent (or product of powers), then power of a power; $4^6 = 4,096$**
- $2.1^2 = 4.41$
- $(\frac{3}{4})^6 = \frac{729}{4096}$
- $\frac{4^{14}}{4^{14}} = 4^0 = 1$
- $3.3 \cdot 3.3^2 = 3.3^3 = 35.937$

6. $(-6)^3 = -216$
7. **A, C, D** *A, C and D*
8. **A** *A*
9. **(a) $n = 6$ (b) $n = 7$**
10. **Yes — power of a quotient, read from right to left; $(-3)^3 = -27$.**

What the audit against Math Nation found

- Math Nation 3.3 pages verified: 1030 (warm-up puzzle maximum), 64, 64, 10000, 729, 0.0625, 4096, 6.25, 64/729, 0.00032, 0.16, 4, -64, 0.0001, 9.261, 0.00001 and the select-all.
- Homework L3 #7 key: 729/4,046 printed; correct 729/4,096. Fixed on the Additional bank.
- Practice L3 word-bank blanks say 'quotient of a power'; the law is 'quotient of powers'. Our item 1 uses the correct names.
- The book's 3.3.4 roundtable is a teams-of-three rotation; its three expressions are whiteboard 2-4.
- The book's warm-up (a digit puzzle: $4^5 + 2 \times 3 = 1030$) is good mathematics but previews nothing; replaced by spaced retrieval with the order-of-operations item in the prior-grade band.

What changed from the book

- 3.3.1 digit-puzzle warm-up → four retrieval questions.
- 3.3.2 Collaboration (Winston/Yvette) → Notes II.
- 3.3.3 Guided Practice (Caroline/Jamir, Ruth) → Example 1 and Example 2.
- 3.3.4 Roundtable and 3.3.5 Your Turn → the nine-question round.
- 3.3.6 Ticket Out the Door → whiteboard 6 (the select-all became a four-option diagnostic) and bank question 8.